KAPASA MAKASA UNIVERSITY

ENTOMOLOGY

UNIT THREE (B)

TOPIC: DESCRIBING INSECT ANATOMY AND PHYSIOLOGY

Objectives: by the end of this unit, SSBAT;

- Explain the external parts of an insect
- Describe the functions of external parts
- Describe internal parts and physiological process.

3.2. Internal parts and physiological process of an insect

3.2.1. Parts and function of the endocrine system

A hormone is a chemical signal sent from cells in one part of an organism to cells in another part (or parts) of the same individual. They are often regarded as chemical messengers. Although typically produced in very small quantities, hormones may cause profound changes in their target cells. Their effect may be stimulatory or inhibitory. In some cases, a single hormone may have multiple targets and cause different effects in each target. There are at least four categories of hormone producing cells in an insect body.

3.2.1.1 Corpora cardiac

These are a pair of neuroglandular bodies that are found behind the brain and on either sides of the aorta. These not only produce their own neurohormones but they store and release other neurohormones including brain hormones, which stimulates the secretory activity of the prothoracic glands, playing and integral role in moulting.

3.2.1.2. Corpora allata

These are small paired glandular bodies originating from the epithelium located on either side of the foregut. They secrete the juvenile hormone which is responsible for regulating reproduction and metamorphosis.

3.2.1.3. Prothoracic glands

Prothoracic glands are diffuse paired glands located at the back of the head. These glands secrete an ecdysteroid called ecdysone or the moulting hormone, which initiates the epidermal moulting process. Additionally it plays a role in accessory reproduction glands in the process of egg production.

3.2.1.4. Neurosecretory cells

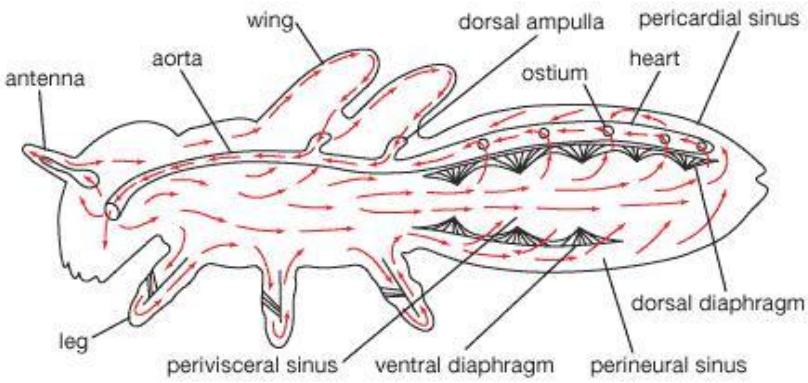
Neurosecretory cells in the brain can produce one or more hormones that affect growth, reproduction, homeosis and metamorphosis. Homeosis is the property of a system in which variables are regulated so that internal conditions remain stable and relatively consistent e.g. regulation of temperature and the balance between acidity and alkalinity (pH).

Circulatory system

The circulatory system in an insect is an open system while the circulatory system of human beings and other vertebrates is completely closed. In human beings blood travels only through special vessels, the arteries, capillaries and veins. The dorsal vessel comprises of the heart and the dorsal aorta is the only blood vessel in insects.

Blood enters the heart through small openings and is pumped forward through the dorsal aorta to the head by peristaltic movements of the heart. And from the head the blood moves freely and slowly through the head, thorax and abdomen irrigating and nourishing all body tissues. The functions of the insect blood are:

1. To transport food, waste products and hormones from one part of the body to another.
2. Lubrication of internal structures
3. To act as a hydraulic medium (e.g. expansion of wings which is done by forcing blood into the wings
4. Protection through phagocytosis (blood cells ingest small foreign materials), encapsulation (blood cells surround the foreign object, coagulation or wound healing).
5. To serve as a simple storehouse for metabolic waste products in many small insects.

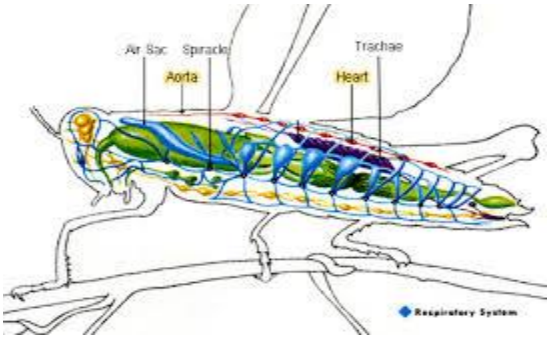


Respiratory system

The intake of oxygen and its transportation to the insect body tissues, as well as the removal of carbon dioxide are done through spiracles and trachea. This is known as breathing system or ventilator system in insects.

The spiracles are external openings of the respiratory system. The spiracles on each side of an insect are connected to a trachea (air tube) that lies longitudinally inside the exoskeleton.

There are two major tracheal trunks, from which the trachea branch in many directions of the insects becoming smaller and numerous, to reach all tissues of the insects body. The fine terminal branches of the trachea are called the tracheoles. The air sacs serve as air storage pockets to aid in respiration.



Excretory system

The excretory organs for most insects are the malphighian tubules.

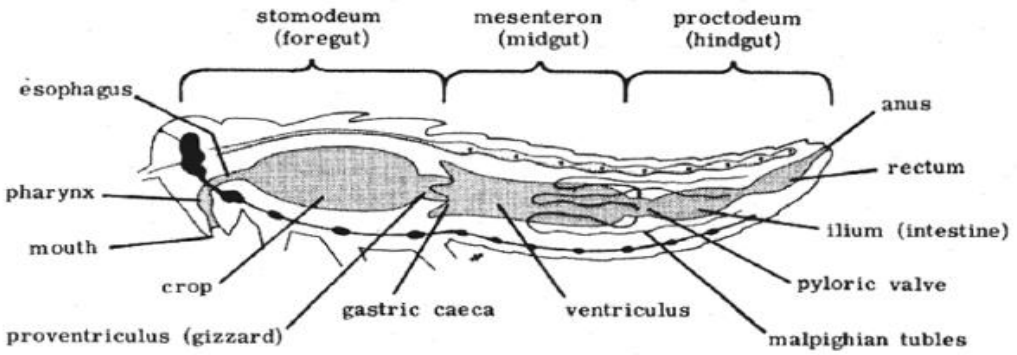
The metabolic waste materials primarily sodium and potassium salts of uric acid are absorbed from the blood by these tubules, transported and then passed out through the hindgut and anus.

The major function of the excretory system is to maintain a constant environment by eliminating waste materials from blood and regulate water and salts.

Digestive system

The various feeding habits of many insects have led to variation in their digestive systems. Digestion and absorption of food occurs in the alimentary canal. The alimentary canal extends from the mouth to the anus and is divided into three parts namely **foregut, midgut** and **hindgut**.

The food is gathered and prepared for digestion by the mouthparts and passed into the alimentary canal. The foregut has the following parts; **Pharynx, oesophagus, crop, ventriculus** and **foregut valve**. The midgut comprises of gastric caeca and the ventriculus (stomach). Gastric caeca are finger like outgrowths located in the anterior proportion of the midgut, produces digestive juice. The **hindgut** is divided into **malphigian tubules, small and large intestines,** and the **rectum.** Malphigian tubules are excretory tubes that originate at the hindgut and branch out in many directions from the alimentary canal. The small and large intestine digest and absorb food that passes from the midgut. The rectum is a sac that accumulates and compresses undigested food that is excreted through the anus. The insect droppings are called **frass.** The rectum re-absorbs water, amino acids and salt. The micro – organisms are present in the digestive system of insects. In termites, the micro-organisms present in the hindgut are important for the digestion of cellulose.

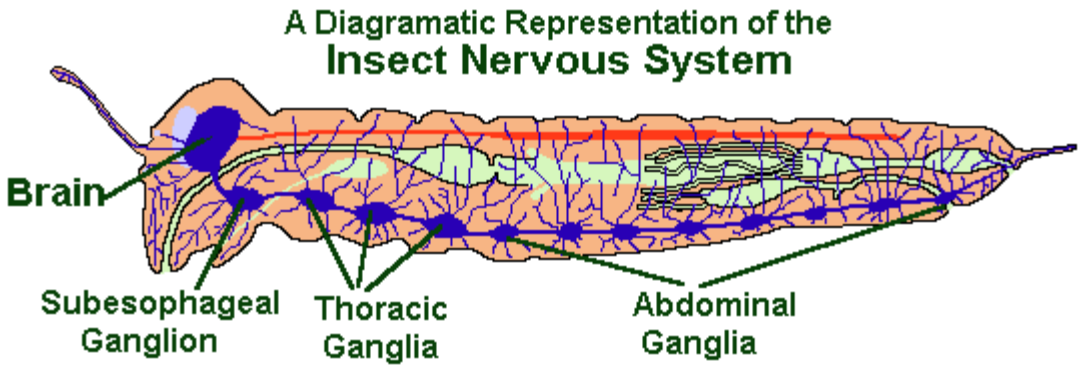


The "generalized" digestive system of insects.

Nervous system

The generalised central nervous system of insects consists of a brain, located dorsally in the head, a pair of connectives around the alimentary canal, and ventrally placed ganglia interconnected to form the ventral nerve cord. Nerves extend from the brain and ventral nerve cord to various body parts. Nerves conduct information from sense organs located on the exoskeleton to the ganglia and flow to muscles.

These sense organs are extremely sophisticated and often are excited by only one specific type of stimuli such as mechanical, chemical, auditory or visual.

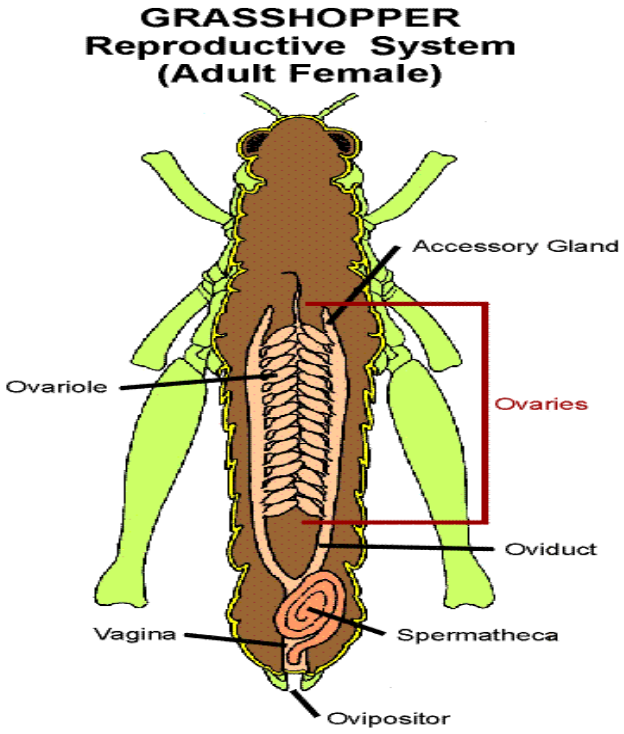


Reproductive system

Most insects reproduce bisexually but some insects reproduce by parthenogenesis in which the egg develops without fertilization. The male reproductive system is located at the posterior end of the abdomen and usually consists of a pair of testes, a number of tubes and accessory glands. The functions of the male system are production, storage and a delivery of the sperm.

The female reproductive system is also located at the posterior end of the abdomen and usually consists of a pair of ovaries, a number of tubes and the vagina. The female reproductive system functions in the production of eggs, storage of eggs and sperm as fertilization site in the oviposition of the fertilized egg.

There are considerable differences in the reproductive system of some insects.



Female and male reproductive parts in insects.

